0647. Palindromic Substrings

Medium · 1-D Dynamic Programming
leetcode.com/problems/palindromic-substrings

Problem

- Count the palindromic substrings of s (different positions count separately).

Intuition

- Same centre expansion as problem 5, but count every successful expansion step.

Brute Force

- Check all substrings: O(n^3).

Optimal Approach

- For every centre (i, i) and (i, i+1): while ends match, count++ and expand.

Worked Example

```
"aaa".

odd centres : a | a, aaa | a          -> 4
even centres: aa | aa                 -> 2
total 6
```

Java Solution

```java
public class PalindromicSubstrings {
    public int countSubstrings(String s) {
        int count = 0;
        for (int i = 0; i < s.length(); i++) count += expand(s, i, i) + expand(s, i, i + 1);
        return count;
    }

    private int expand(String s, int l, int r) {
        int c = 0;
        while (l >= 0 && r < s.length() && s.charAt(l) == s.charAt(r)) { c++; l--; r++; }
        return c;
    }
}
```

Complexity

- Time: O(n^2). Space: O(1).

Pitfalls

- Single characters count.